Charge

1. Conservation – isolated system total electric charge
2. Quantization

   smallest unit of charge e → elementary
   ↳ proton or electron

   q ← variable

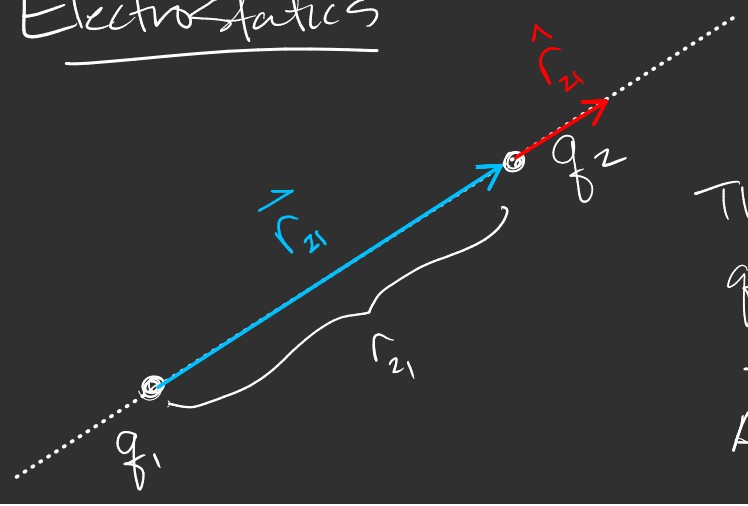
SI units → Coulomb $\quad 1e \equiv 1.602 \cdot 10^{-19}\,C$

$$\downarrow$$

$$1C = \frac{1}{1.602 \cdot 10^{-19}}\,e = 6.242 \cdot 10^{18}\,e$$

Electrostatics

$$\hat{a} = \frac{\vec{a}}{|\vec{a}|} \qquad r^2 = \vec{r} \cdot \vec{r}$$



The force between q_1 + q_2 is directly proportional to the product of q_1 + q_2 AND inversely proportional to distance squared btt them

Coulomb's Law

$$\vec{F}_{21} = \frac{k\, q_1 q_2}{r_{21}^2}\,\hat{r}_{21}$$

Coulomb constant

electric constant

$$k = 8.988 \cdot 10^{9}\ \frac{N\,m^2}{C^2} \approx 9 \cdot 10^{9}\ \frac{N\,m^2}{C^2}$$

$$k = \frac{1}{4\pi\epsilon_0}$$

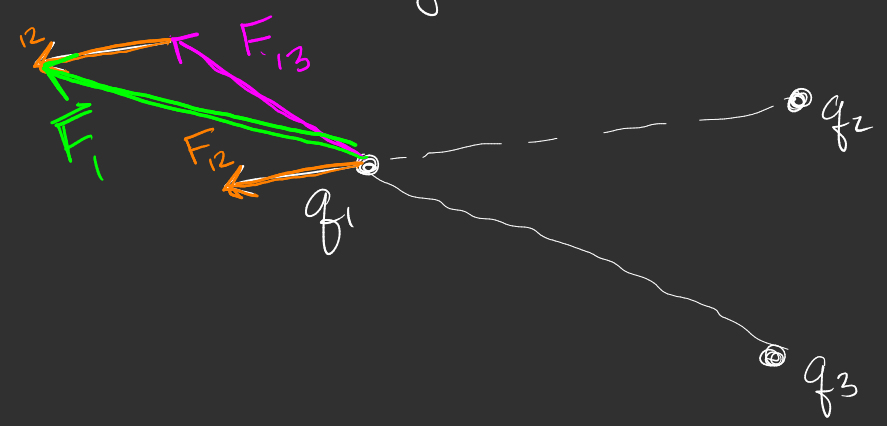
permittivity of free space

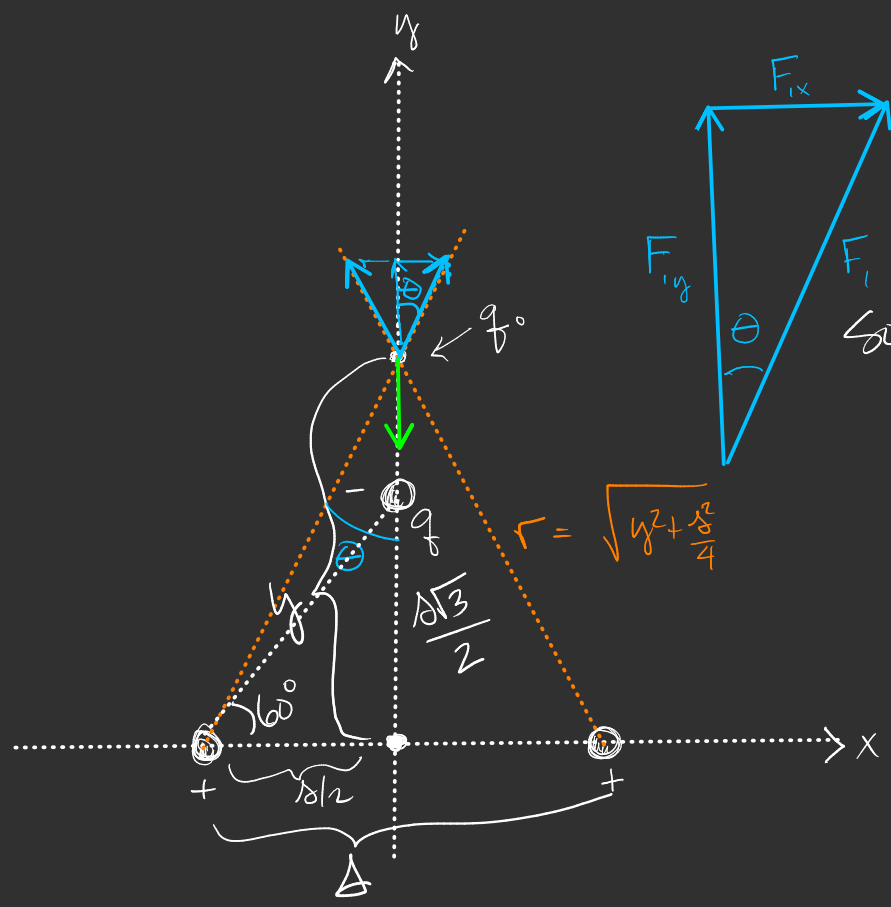
$$\epsilon_0 = 8.85 \cdot 10^{-12}\ \frac{C^2}{N\,m^2}$$

What about more than two charges?

→ Superposition

$$\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13}$$





SOHCAHTOA

$$\cos\theta = \frac{y}{\sqrt{y^2 + \frac{\delta^2}{4}}}$$

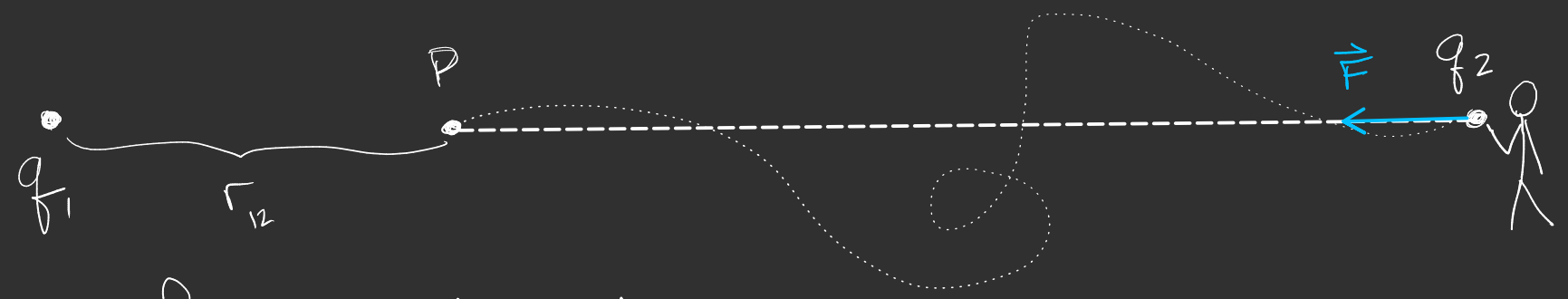
$$0 = F = 2 \cdot \frac{k\cancel{q}q_0}{y^2 + \frac{\delta^2}{4}} \cdot \frac{y}{\sqrt{y^2 + \frac{\delta^2}{4}}} - \frac{k\cancel{q}q_0}{\left(y - \frac{\delta\sqrt{3}}{2}\right)^2}$$

$$\frac{1}{\left(y - \frac{\delta\sqrt{3}}{2}\right)^2} = \frac{2 \cdot y}{\left(y^2 + \frac{\delta^2}{4}\right)^{3/2}}$$

$$y = \frac{\left(y^2 + \frac{\delta^2}{4}\right)^{3/2}}{2\left(y - \frac{\delta\sqrt{3}}{2}\right)^2}$$

$\delta = 1$ $\quad y = 3.1, 0.29, 0.12$?

Work + Energy



$$W = \int \text{force} \cdot \text{displacement}$$

$$W = \int \vec{F} \cdot d\vec{r}$$

Ex. move radially

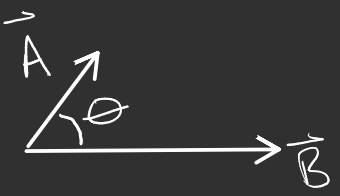
$$W_{ME} = \int_{\infty}^{r_{12}} -\frac{kq_1q_2}{r^2}\,\hat{r} \cdot d\vec{r}$$

$$= -kq_1q_2 \int_{\infty}^{r_{12}} r^{-2}\,dr$$

$$= -kq_1q_2 \left(\frac{(-1)}{r}\right)\Bigg|_{\infty}^{r_{12}}$$

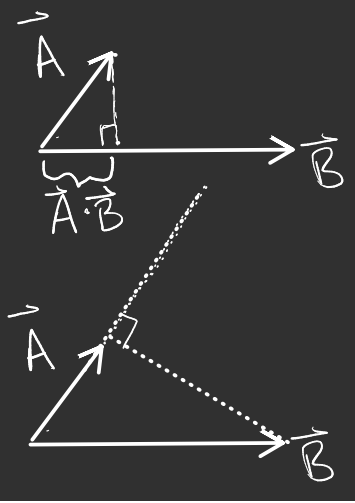
$$\boxed{W_{ME} = \frac{kq_1q_2}{r_{12}}}$$

$$\vec{A} \cdot \vec{B} = |A||B| \cos\theta$$

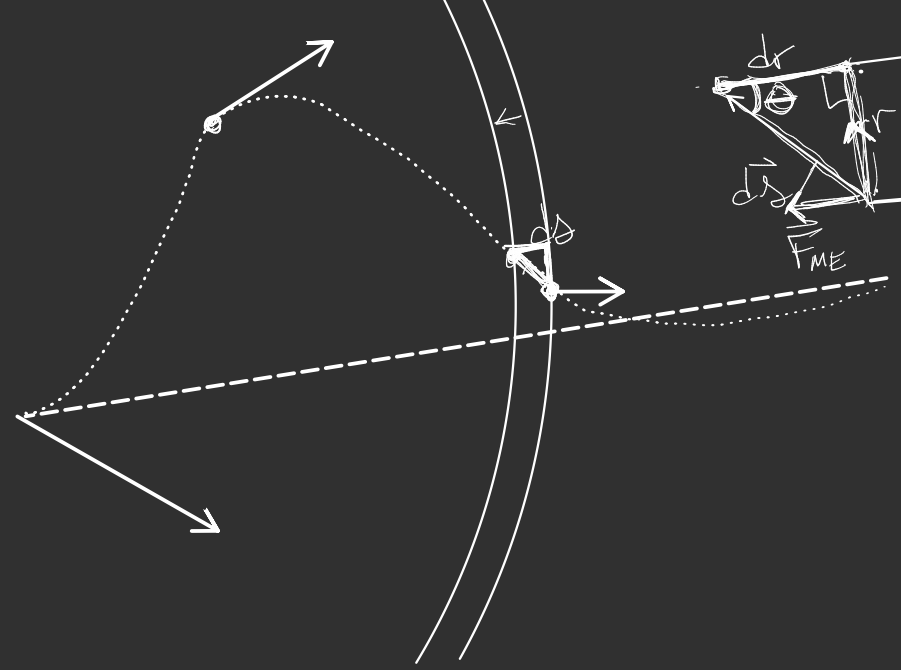


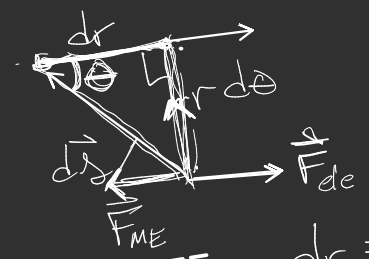
$$\vec{A} = (A_x\hat{x} + A_y\hat{y} + A_z\hat{z})(B_x\hat{x} + B_y\hat{y} \ldots$$

$$\vec{A} \cdot \vec{B} = A_xB_x + A_yB_y + A_zB_z$$



$$\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$$





$$dW = \vec{F} \cdot d\vec{s}$$

$$dr = ds\cos\theta$$

$$\frac{dr}{\cos\theta} = ds$$

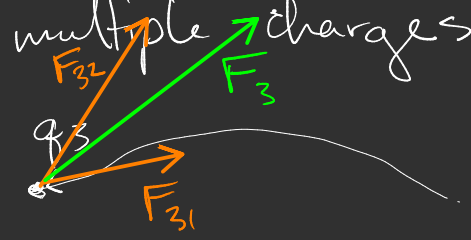
$$\vec{F} \cdot d\vec{s}$$

$$F\,ds\cos\theta$$

$$F\,dr$$

$$\vec{\nabla} \cdot \vec{F} = 0$$

What about multiple charges?



q_1

q_2

$$\vec{F}_3 = \vec{F}_{32} + \vec{F}_{31}$$

$$W_3 = \int \vec{F}_3 \cdot d\vec{r}$$

$$= \int \left(\vec{F}_{32} + \vec{F}_{31}\right) \cdot d\vec{r}$$

$$= \int \vec{F}_{32} \cdot d\vec{r} + \int \vec{F}_{31} \cdot d\vec{r}$$

$$W_3 = W_{32} + W_{31}$$

So what is the total work?

$$W_T = W_{21} + W_{31} + W_{32}$$

$$W_T = k\left(\frac{q_1 q_2}{r_{21}} + \frac{q_1 q_3}{r_{31}} + \frac{q_2 q_3}{r_{32}}\right)$$

Since F_e is conservative, W to build the charges → potential energy

Remington: $W_{ext} = U_{electric}$

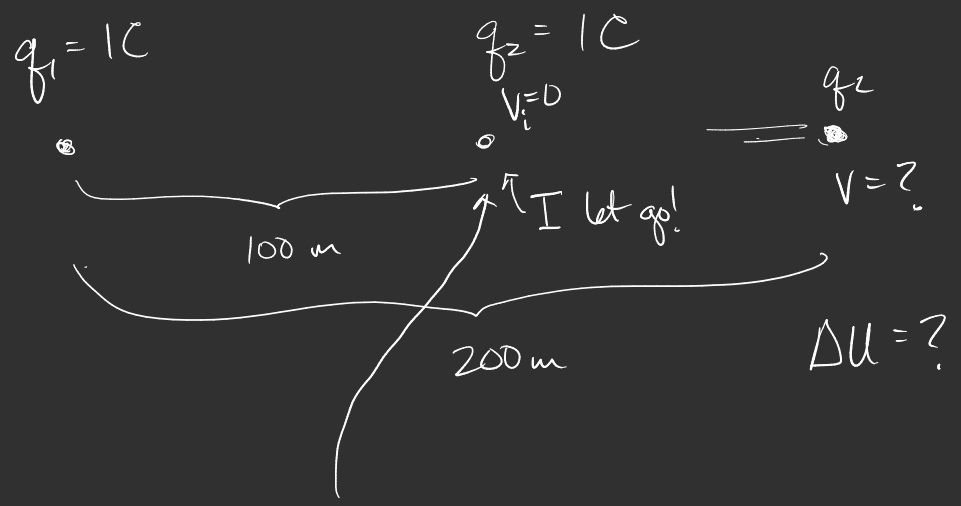
Books: $W_{charges} = -U_{electric}$

conservation of energy →

$$E_i + \underbrace{\Delta E}_{W_{ext}} = \underbrace{E_f}_{K_f + U_f}$$

$$K_i + U_i + \underbrace{W_{ext}}_{\text{non conservative}} = K_f + U_f$$

$$W_{ext} = \Delta K + \Delta U$$



$$W_{ext} = 0$$

$$0 = \Delta K + \Delta U$$

$$\boxed{\Delta K = -\Delta U}$$

$$U(r) = W(\infty \to r)$$

$$U(r) = \frac{k q_1 q_2}{r}$$

$$\Delta U = U(200\,m) - U(100\,m)$$

$$= \frac{9 \cdot 10^9 \cdot 1C \cdot 1C}{200} - \frac{9 \cdot 10^9 \cdot 1C \cdot 1C}{100}$$

$$= 9 \cdot 10^9 \left(\frac{1}{200} - \frac{1}{100} \right)$$

$$= -4.5 \cdot 10^9\ J$$

$$K_i = 0$$

$$\Delta K = 4.5 \cdot 10^9\ J \rightsquigarrow K_f = \frac{1}{2} m v_f^2$$

$$\Delta K = K_f - K_i$$

$$= \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

$$= \frac{1}{2} m \left(v_f^2 - v_i^2 \right)$$

Electric Field

$$\vec{F} = q_0 \vec{E}$$

$$\vec{E} = \frac{\vec{F}}{q}$$

force per unit charge.

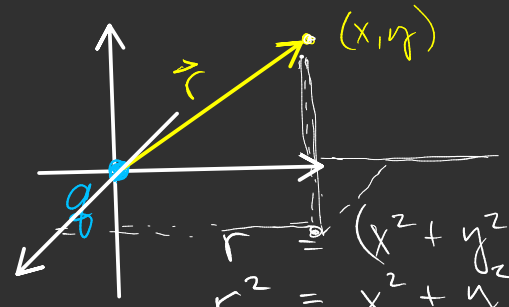
for a point charge

$$\vec{E}(\vec{r}) = \frac{kq}{r^2}\hat{r}$$

state in cartesian coords.

source at origin

position vector $\rightarrow$ $\vec{r} = x\hat{x} + y\hat{y} + z\hat{z}$



$$r = (x^2 + y^2 + z^2)^{1/2}$$

$$r^2 = x^2 + y^2 + z^2$$

$$\hat{r} = \frac{x\hat{x} + y\hat{y} + z\hat{z}}{(x^2 + y^2 + z^2)^{1/2}}$$

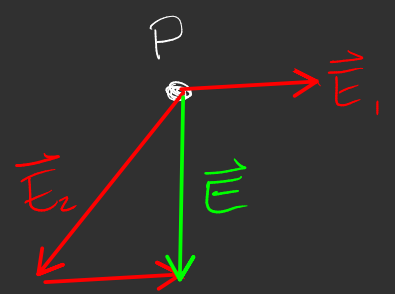
for a charge distribution

$$\vec{E}(x,y,z) = \frac{1}{4\pi\epsilon_0}\sum_{j=1}^{N}\frac{q_j}{r_{0j}^2}\hat{r}_{0j}$$

$$\vec{E}(x,y,z) = \frac{kq\cdot(x\hat{x} + y\hat{y} + z\hat{z})}{(x^2 + y^2 + z^2)(x^2 + y^2 + z^2)^{1/2}}$$

$$= \frac{kq\cdot(x\hat{x} + y\hat{y} + z\hat{z})}{(x^2 + y^2 + z^2)^{3/2}}$$

$q_1 = 1C$



$q_2 = -2C$

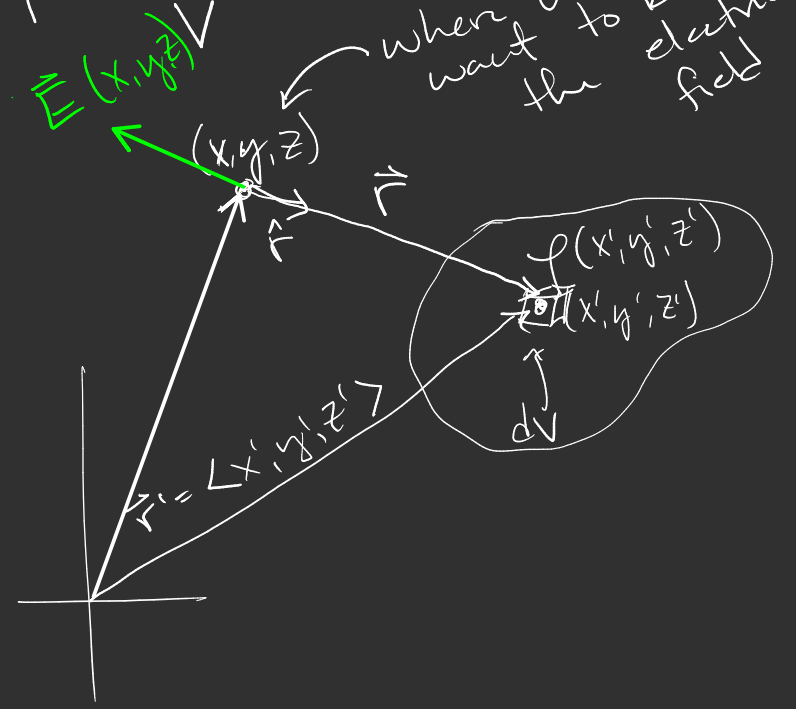
Continuous charge distribution

$$\rho \longrightarrow \rho(x', y', z') \qquad dq = \rho(x', y', z')\, dV$$

volume element

charge density
charge per volume

$$\rho = \frac{Q}{V}$$

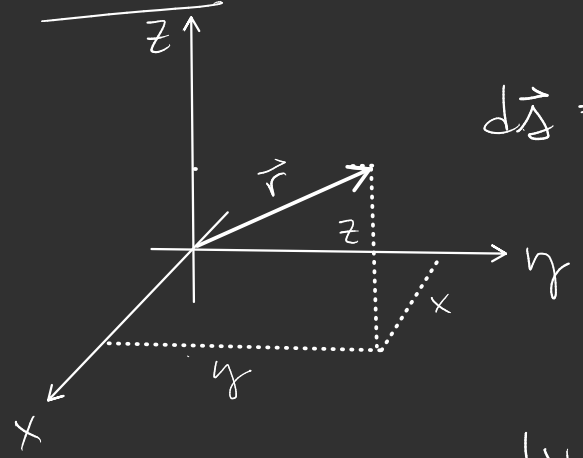


$$\vec{E}(x,y,z) = \frac{1}{4\pi\epsilon_0} \sum_{j=1}^{N} \frac{q_j \hat{r}_{0j}}{r_{0j}^2}$$

$$\Downarrow$$

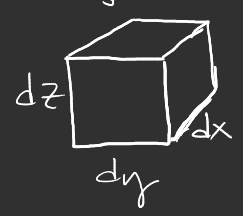
$$\vec{E}(x,y,z) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(x',y',z')\hat{r}\, dV'}{r^2}$$

cartesian:



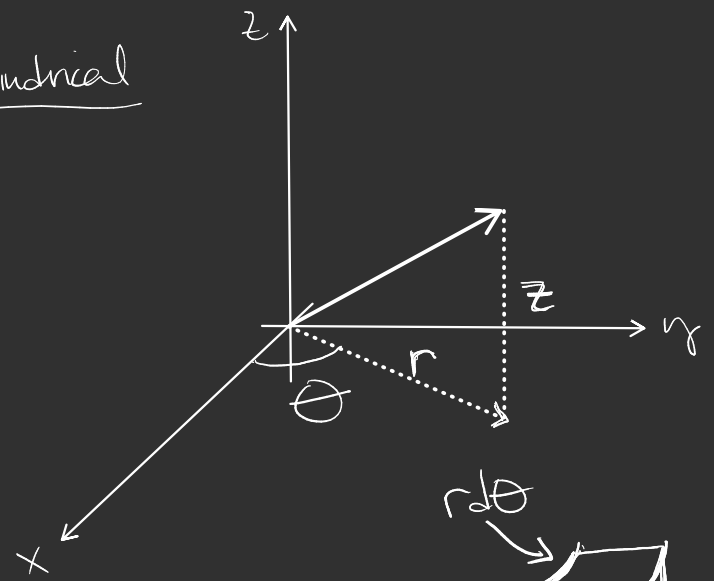
$$d\vec{l} = dx\,\hat{x} + dy\,\hat{y} + dz\,\hat{z}$$

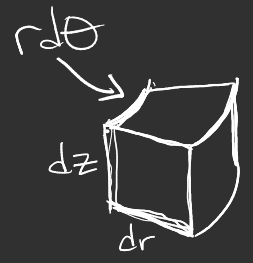
cube



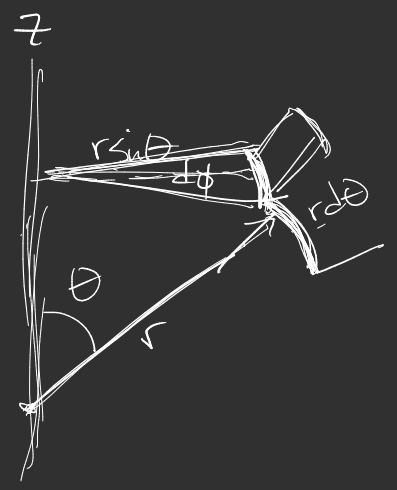
$$dV = dx\,dy\,dz$$

cylindrical

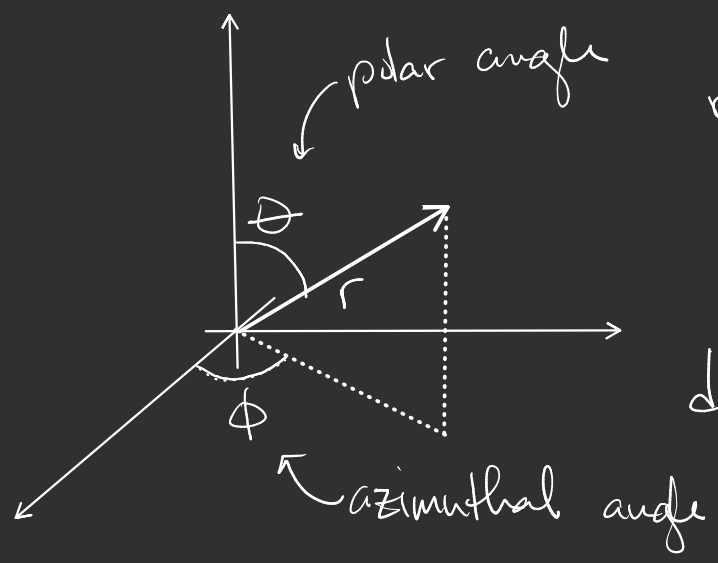


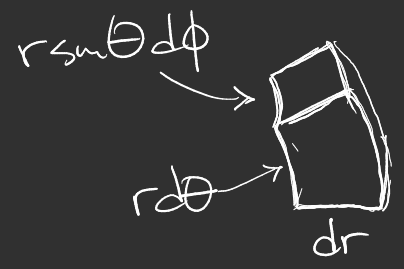


$$dV = r\,dr\,d\theta\,dz$$

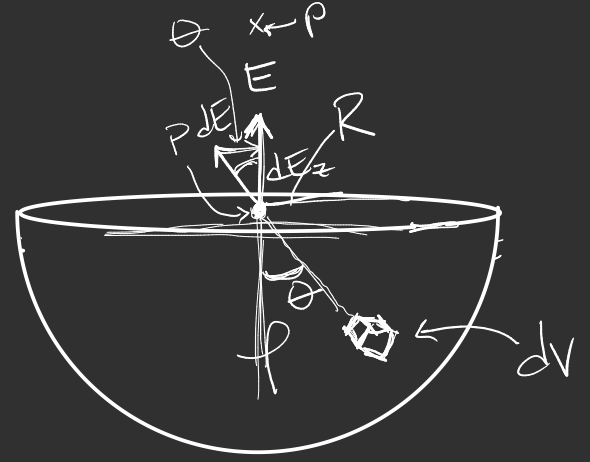


spherical (polar)





$$dV = r^2 \sin\theta\,dr\,d\theta\,d\phi$$



uniform charge density $\rightarrow \rho$

$$E = \frac{1}{4\pi\epsilon_0}\int \frac{\rho\, dV}{r^2}$$

$$\Downarrow$$

$$dE_z = \frac{\rho\, dV}{4\pi\epsilon_0 r^2}\cos\theta$$

$$dE_z = \frac{\rho\, \cancel{r^2}\sin\theta\cos\theta\, dr\, d\theta\, d\phi}{4\pi\epsilon_0 \cancel{r^2}}$$

$$E_z = \frac{\rho}{4\pi\epsilon_0}\int_{r=0}^{R}\int_{\theta=0}^{\pi/2}\int_{\phi=0}^{2\pi} \sin\theta\cos\theta\, dr\, d\theta\, d\phi$$

$$E_z = \frac{\rho R}{\cancel{4}_{2}\pi\epsilon_0}\cancel{(2\pi)}\underbrace{\int_0^{\pi/2}\sin\theta\cos\theta\, d\theta}_{1/2}$$

$$\left(\sin\theta\cos\theta = \frac{1}{2}\sin 2\theta\right)$$

$$\boxed{E_z = \frac{\rho R}{4\epsilon_0}}$$

Flux → electric flux

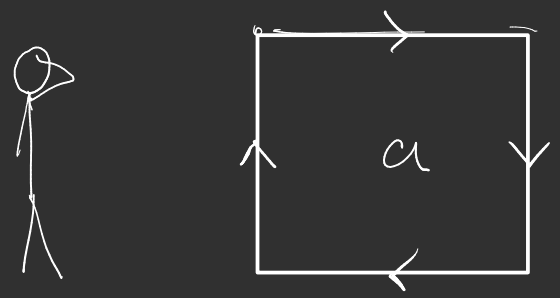
$dy = \frac{dy}{dx}\,dx$

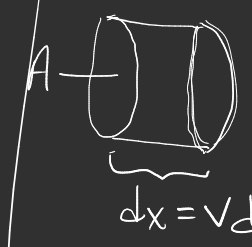
↳ electric field that "passes through" an area

$$\Phi = \vec{E} \cdot \vec{a}$$

dot product

$$d\Phi = \vec{E} \cdot d\vec{a}$$

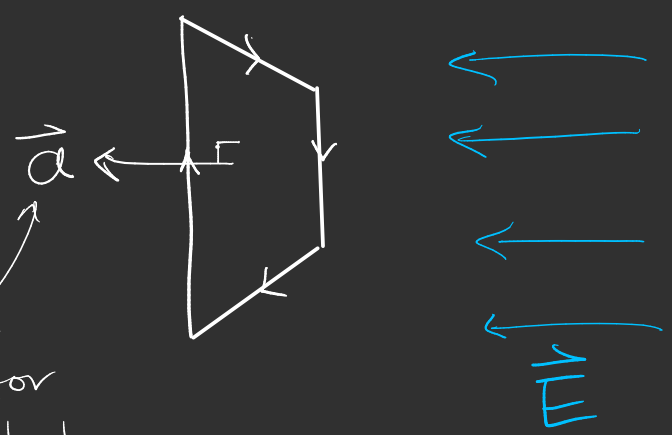




$$\frac{dV}{dt} = (A \cdot \frac{v}{dt})dt$$

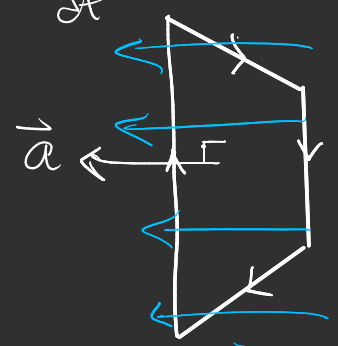
$$\frac{dV}{dt} = \vec{A} \cdot \vec{v}$$

square

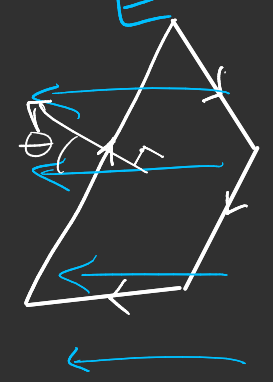


area vector defined by RHR

$$\Phi = |\vec{E}||\vec{a}|\cos 0^\circ$$



$$\phi = |\vec{E}||\vec{a}|\cos\theta$$

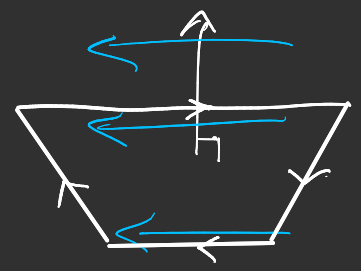


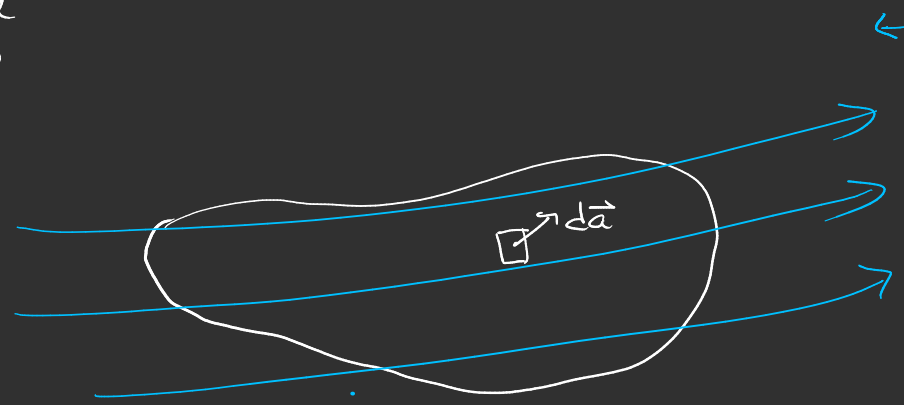
Gauss's Law

Net Flux through a closed surface is proportional to the charge inside the surface.

$$\Phi_{NET} = \frac{q_{encl}}{\epsilon_0}$$

$\phi = 0$

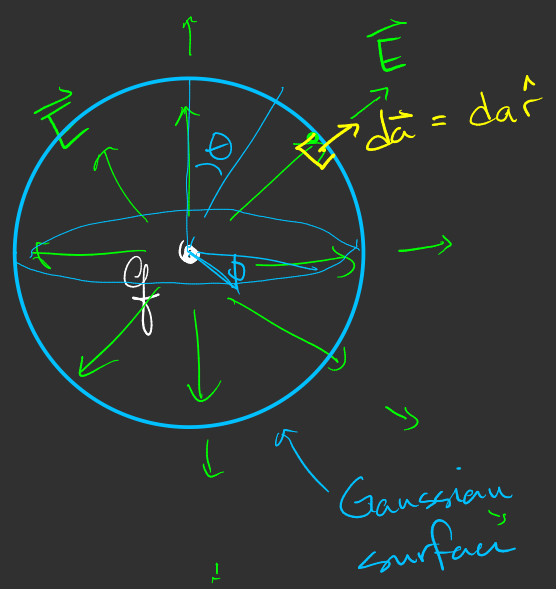




$$d\Phi = \vec{E} \cdot d\vec{a}$$

$$\Phi = \int_S \vec{E} \cdot d\vec{a}$$

Let's check w/ a point charge



$$\vec{E}(\vec{r}) = \frac{q_0}{4\pi\epsilon_0 r^2}\hat{r} \leftarrow \text{spherical}$$

$$\vec{E} \cdot d\vec{a} = \frac{q_0}{4\pi\epsilon_0 r^2}\, da\, \underbrace{\hat{r} \cdot \hat{r}}_{1}$$

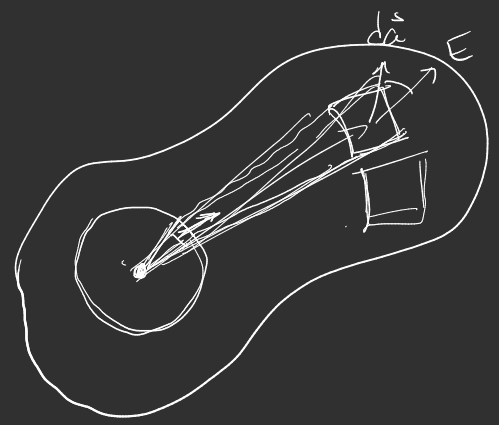
$$\Phi = \int_S \frac{q_0}{4\pi\epsilon_0 r^2}\, da$$

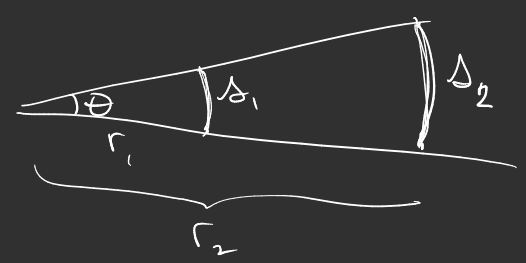
thorough way

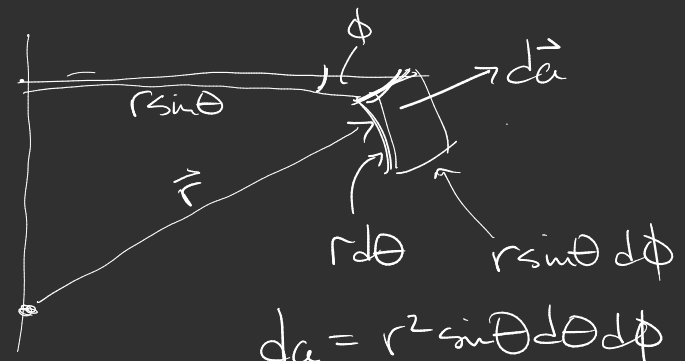
$$\Phi = \frac{q_0}{4\pi\epsilon_0 r^2} \underbrace{\int_S da}_{4\pi r^2}$$

$$\Phi = \frac{q_0 \, (4\pi r^2)}{4\pi\epsilon_0 r^2}$$

$$\Phi = \frac{q_0}{\epsilon_0} \quad \checkmark\checkmark$$







$$da = r^2 \sin\theta \, d\theta \, d\phi$$

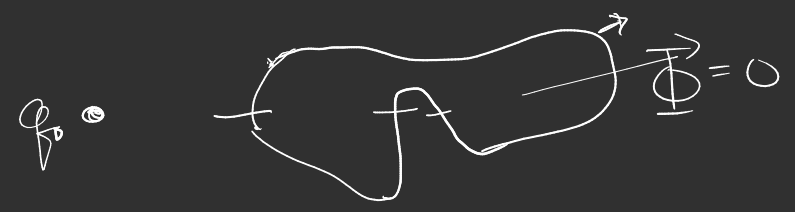
$$\Phi = \frac{q_0}{4\pi\epsilon_0} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{1}{r^2} \cdot r^2 \sin\theta \, d\theta \, d\phi$$

$$\Phi = \frac{q_0}{4\pi\epsilon_0} \cdot \left[-\cos\theta\right]_0^{\pi} \cdot 2\pi$$

$$-(-1)-(-1) = 2$$

$$\Phi = \frac{q_0 \cdot 2 \cdot 2\pi}{4\pi\epsilon_0}$$

$$\Phi = \frac{q_0}{\epsilon_0}$$



$$\Phi = \int_S \vec{E} \cdot d\vec{a} = \int (\vec{E}_1 + \vec{E}_2 + \dots) d\vec{a} = \frac{q_{encl.}}{\epsilon_0}$$

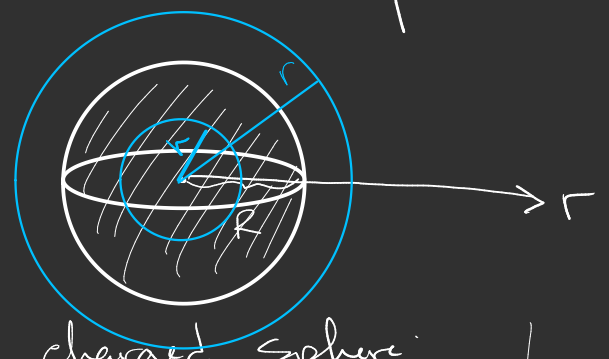
$$q_{encl.} = \int_{\substack{\text{charged} \\ \text{volume} \\ \text{enclosed}}} \rho \, dV$$

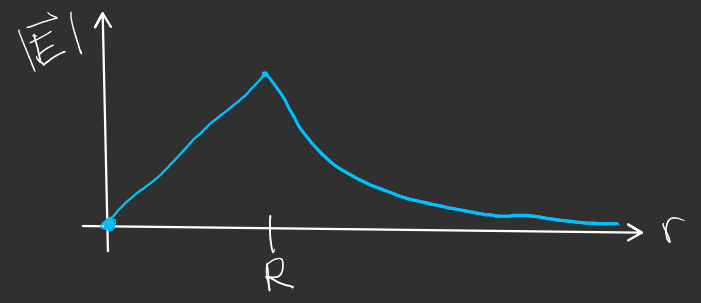
$$\int_S E \cdot d\vec{a} = \frac{1}{\epsilon_0} \int \rho \, dV_{encl}$$ ← Gauss's Law

1. Electric field for a spherical charge distribution inside and outside the sphere
2. Electric field for an infinite line charge
3. Electric field for an infinite plane of charge.

$\rho \propto \frac{1}{r^2}$

ρ is uniform





inside the charged sphere: $(r < R)$

flux

$$\int_S (\vec{E} \cdot d\vec{a}) = \frac{1}{\epsilon_0} \int \rho \, dV_{encl}$$

$$E \cdot 4\pi r^2 = \frac{1}{\epsilon_0} \rho \underbrace{\int dV_{encl}}_{\frac{4}{3}\pi r^3}$$

$$E \cdot \cancel{4\pi} r^{\cancel{2}} = \frac{\cancel{4\pi} \rho r^{\cancel{3}}}{3\epsilon_0}$$

$$\boxed{E = \frac{\rho r}{3\epsilon_0}} \quad r < R$$

outside the charged sphere: $r > R$

$$\int_S \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} \int \rho \, dV_{encl}$$

$$E \cdot 4\pi r^2 = \frac{\rho}{\epsilon_0} \underbrace{\int dV_{encl}}_{\frac{4}{3}\pi R^3}$$

$Q_{encl.} = \rho V$

$$E \cdot \cancel{4\pi} r^2 = \frac{\cancel{4\pi} \rho R^3}{3\epsilon_0}$$

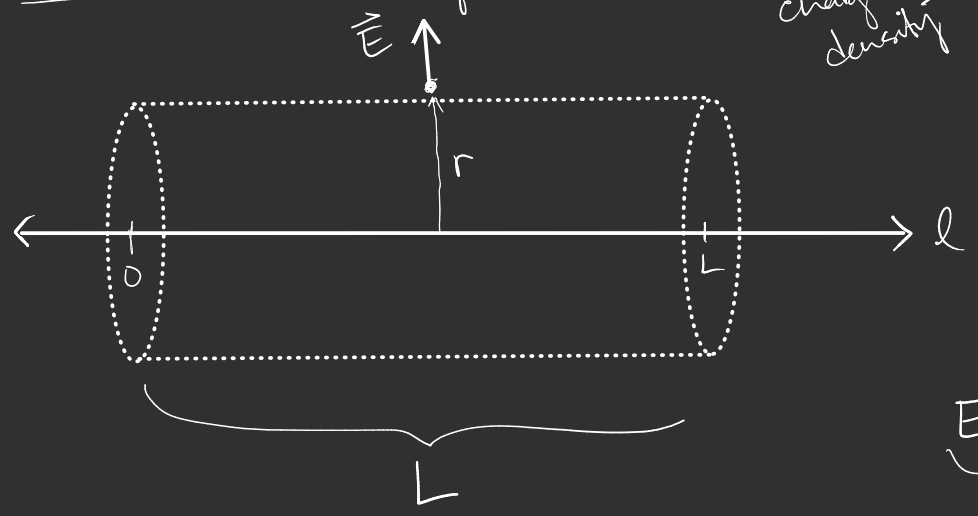
$$E = \frac{\rho R^3}{3\epsilon_0 r^2}$$

$$E \cdot 4\pi r^2 = \frac{Q}{\epsilon_0}$$

$$\boxed{E = \frac{Q}{4\pi\epsilon_0 r^2}}$$

$r > R$

infinite line charge



linear charge density $\lambda = \frac{dq}{dl}$

$$dq = \lambda \cdot dl$$

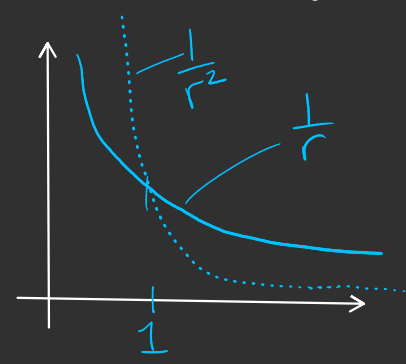
$$\int \vec{E} \cdot d\vec{a}$$

$$\underbrace{E \cdot L \cdot 2\pi r}_{\text{sides}} + \underbrace{0}_{\text{ends}} = \frac{1}{\epsilon_0} \int \lambda \cdot dl_{encl}$$

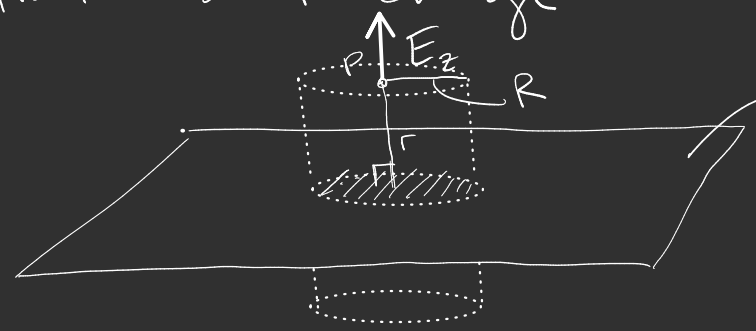
$\vec{E} \cdot d\vec{a}_{end} = 0$

$$E \cdot \cancel{L} \cdot 2\pi r = \frac{\lambda \cdot \cancel{L}}{\epsilon_0}$$

$$E = \frac{\lambda}{2\pi \epsilon_0 r}$$



infinite sheet charge



area charge density

$$\sigma = \frac{dq}{da}$$

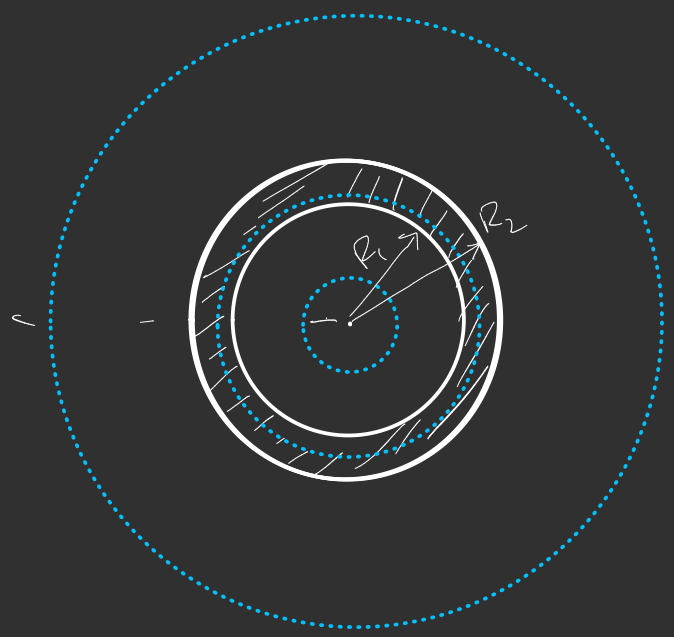
$$\int \vec{E} \cdot d\vec{a} = \frac{1}{\epsilon_0} \int \sigma \, da_{end}$$

$$\underbrace{E \cdot \pi R^2}_{\text{top}} + \underbrace{E \cdot \pi R^2}_{\text{bottom}} + \underbrace{0}_{\text{sides}} = \frac{\sigma}{\epsilon_0} \pi R^2$$

$$E \cdot 2\cancel{\pi R^2} = \frac{\sigma}{\epsilon_0} \cancel{\pi R^2}$$

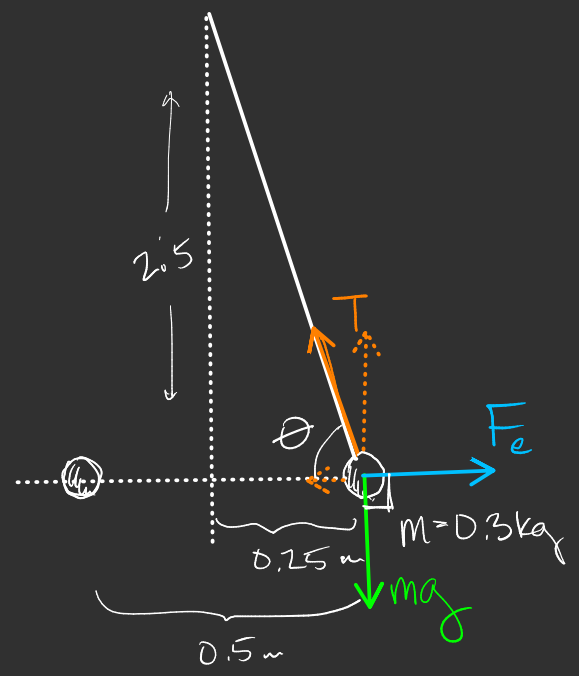
$$\boxed{E = \frac{\sigma}{2\epsilon_0}}$$

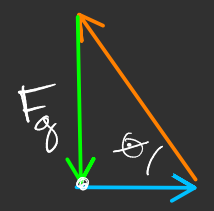
independent of r!?



Homework

36)





$$\sin\theta = \frac{F_g}{T} \qquad \cos\theta = \frac{F_e}{T}$$

$T =$

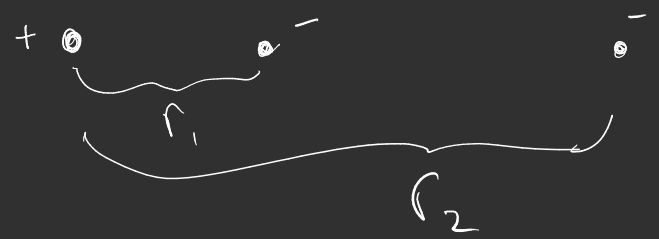
$F_e \rightarrow q$

$$\tan^{-1}\left(\frac{2.5}{0.25}\right)$$

$$T_y \Rightarrow T\sin\theta - mg = 0$$

$$\sin\theta = \frac{mg}{T}$$

40)



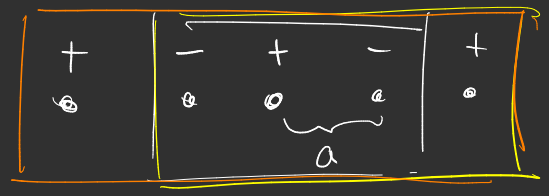
$$-\frac{\cancel{ke^2}}{r_1} - \frac{\cancel{ke^2}}{r_2} + \frac{\cancel{ke^2}}{r_2 - r_1} = 0$$

$$\frac{1}{r_2 - r_1} = \frac{1}{r_1} + \frac{1}{r_2}$$

$$= \frac{r_2}{r_1 r_2} + \frac{r_1}{r_1 r_2} = \frac{r_2 + r_1}{r_1 r_2}$$

$$r_2 - r_1 = \frac{r_1 r_2}{(r_2 + r_1)}$$

42



$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots\cdots$$

$$-\frac{ke^2}{a} - \frac{ke^2}{a} + \frac{ke^2}{2a}$$

$$\boxed{-\frac{2ke^2}{a} + \frac{ke^2}{2a} - \frac{ke^2}{a} + \frac{ke^2}{2a} - \frac{ke^2}{3a}}$$

$$-\frac{3ke^2}{a} + \frac{2ke^2}{2a} - \frac{ke^2}{3a} - \frac{ke^2}{a} + \frac{ke^2}{2a} - \frac{ke^2}{3a} + \frac{ke^2}{4a}$$

$$-\frac{4ke^2}{a} + \frac{3ke^2}{2a} - \frac{2ke^2}{3a} + \frac{ke^2}{4a}$$

$$\frac{U}{N} = -\frac{ke^2}{N}\left(\frac{(N-1)}{a} - \frac{N-2}{2a} + \frac{N-3}{3a} - \frac{N-4}{4a} + \cdots\right)$$

$$(r_2 + r_1)(r_2 - r_1) = r_1 r_2$$

$$\frac{r_2^2 - r_1^2}{r_1^2} = \frac{r_1 r_2}{r_1^2}$$

$$\frac{r_2^2}{r_1^2} - 1 = \frac{r_2}{r_1}$$

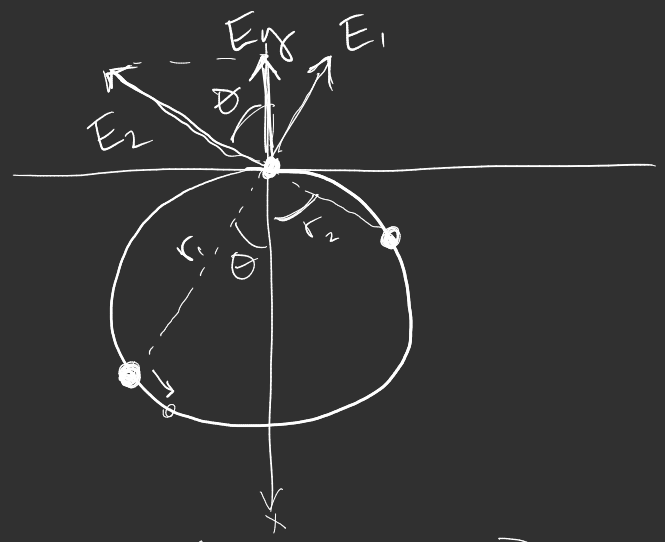
$$\left(\frac{r_2}{r_1}\right)^2 - 1 = \frac{r_2}{r_1}$$

$$\left(\frac{r_2}{r_1}\right)^2 = \frac{r_2}{r_1} + 1$$

$$x^2 = x + 1$$

$$1.6180\cdots\cdots$$

49



$$E_y = \frac{kq}{r^2}\cos\theta$$

want to be constant

$$\frac{\cos\theta}{r^2} = A$$

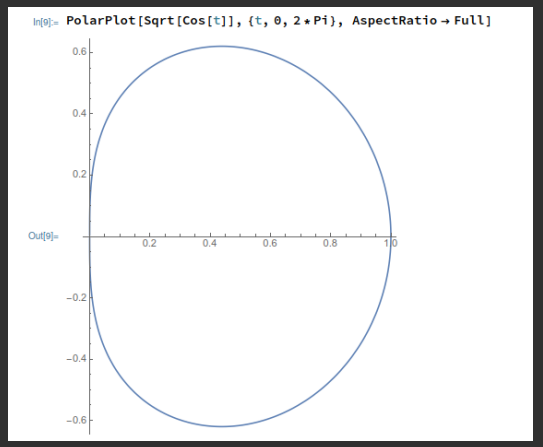
$$r^2 = B\cos\theta$$

$$r = \sqrt{B\cos\theta}$$

$$r = C\sqrt{\cos\theta}$$

Plot!

```
In[9]:= PolarPlot[Sqrt[Cos[t]], {t, 0, 2*Pi}, AspectRatio → Full]
```



$$r^2 = x^2 + y^2$$

$$\cos\theta = \frac{x}{\sqrt{x^2+y^2}}$$

change to cartesian

$$x^2 + y^2 = \frac{Bx}{(x^2+y^2)^{1/2}}$$

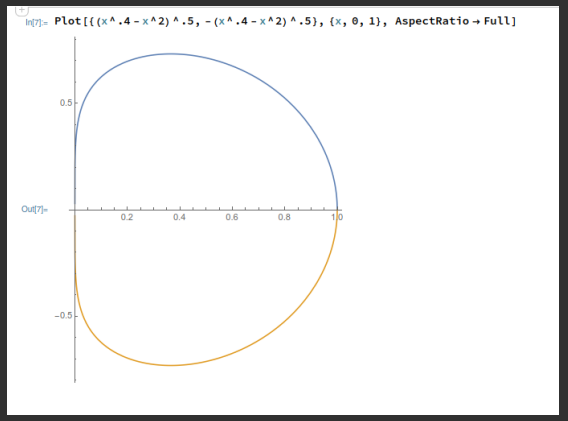
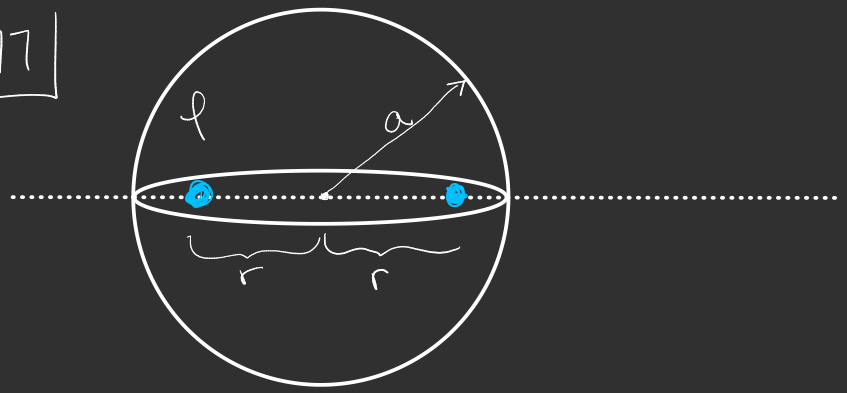
$$(x^2+y^2)^{3/2} = Bx$$

$$x^2 + y^2 = Cx^{2/3}$$

$$y = \left(Cx^{2/3} + x^2\right)^{1/2}$$

Plot! →

```
In[7]:= Plot[{(x^.4 - x^2)^.5, -(x^.4 - x^2)^.5}, {x, 0, 1}, AspectRatio → Full]
```



77